J-1153 · REIS MAGOS GOA HOUSING

From the consultant's CAD to a Revit model, automatically

What the structural CAD already gives us, what is blocking a script, and the drawing standard that closes the gap.

Wing C · 1st floor plan GFC dated 14-07-2026 Reviewed 24-09-2026

THE ANSWER, UP FRONT

The drawings are close. Three things stand in the way.

01

The DWG is already machine-readable

Columns, beams and girders sit on their own layers, and 121 beam tags, 48 column labels and 65 slab tags are real text a script can read.

02

Three gaps block the script

No closed slab outlines, no grid or shared base point, and beam lines drawn on the beam edge instead of its centreline.

03

The fix is a drawing standard

Not new software. A layer list, seven drawing rules and a submission checklist, agreed once and applied to every floor.

WHY IT MATTERS

Today the model is redrawn. Every revision redraws it again.

Manual, as we work now

- Read each size off the printed schedule, mark by mark
- Place 48 columns, 170 beam segments and 65 slab panels by hand, per floor
- Re-type reinforcement into Revit, or leave it out of the model
- A GFC revision repeats most of that work

Automated, once the CAD is right

- The script reads the DWG layers and the Excel schedules
- Geometry, marks, sizes, grade and steel land in one pass
- A revision becomes a re-run and a difference report
- An audit list names every mark it could not place

Effort today [__ hrs per floor] · target [__ hrs] — to be filled from our Wing C record

THE SCALE OF ONE FLOOR

Wing C, first floor, counted from the DWG

48

column labels on the plan, 38 matching the schedule size exactly

170

beam segments drawn, as multilines, plain lines and polylines

65

slab tags, against zero closed outlines to build floors from

21

issues in the audit, every one of them fixable in the drawing

One floor of one wing. J-1153 runs to [__] floors across [__] wings.

HOW IT WORKS

Five steps from DWG to a checked Revit model

1

Read the layers

Column, wall, beam, girder, slab and sunk layers come out as shapes with coordinates.

2

Match every tag to its shape

Each mark text is joined to the nearest element, so C-3 and B-34 attach to real geometry.

3

Join the schedules

Excel rows supply size, thickness, concrete grade and reinforcement for each mark.

4

Place in Revit

Levels and grids first, then columns, beams, floors with their sunk offsets, then the 230 walls.

5

Audit against the sheet

Marks without geometry, geometry without marks, clashes, and Revit schedules compared to the GFC tables.

Six things the drawing already does right

Layers are already split

COLUMN, FL COL, BEAM, GIRDER BEAM, SLAB NO and the sunk hatches each have their own layer.

Columns are closed shapes

Most are closed rectangles, which give a centre point, a size and a rotation directly.

Beams sit at one width

113 beams are 230 wide multilines, so their lines can be pulled out as they are.

Tags are real text

121 beam tags, 48 column labels and 65 slab tags, all readable rather than exploded.

Sizes agree with the schedule

38 of the 48 column labels match the scheduled size exactly, with no interpretation.

The schedules are complete

Beams, GB-1 to GB-6, C-1 to C-32, FC-1 to FC-6 and S-1 to S-10, with concrete grade.

THE THREE THAT STOP A SCRIPT DEAD

Everything else is tidying. These three are structural.

Blocker 01

Slabs have no outlines

The SLAB layer holds six open fragments. There are 65 slab tags and not one closed shape to turn into a Revit floor.

Needed: one closed outline per slab panel.

Blocker 02

Nothing to set out from

No grids and no dimensions in the plan area. The drawing sits about 2.3 km from its origin, with no point shared with the architectural model.

Needed: a grid layer and one agreed base point.

Blocker 03

Beam lines are faces

The multilines are top justified, so every drawn line is a beam face, 115 mm off centre, and nothing says which side.

Needed: one centreline per span.

Columns: duplicates, open shapes and stray heights

Issue	What the DWG contains	What closes it
Duplicates	10 column outlines drawn twice, one over the other	Delete duplicates before issue
Open outlines	C-24, C-25, C-30 and two unlabelled shapes are open 5-point polylines	Close every column outline
C-30 size	Drawn 230x827; the schedule says 230x950; an unlabelled 230x950 sits 0.9 m away	Confirm the size and retag
Not in schedule	C-11, 15, 16, 19 and 20 are drawn at 350x750	Confirm as column stops
Unlabelled shapes	350x1480, 350x2000, 230x230, plus lift-wall pieces 230x2110 and 230x1950	Tag, or move to the wall layer
Stray Z heights	C-3A, 4A, 7A and 10A sit at Z = -11,425; tags range from 229 to 14,784	Flatten the drawing to Z = 0
Tags too far	Column numbers sit 200 to 1,400 mm away from the column they label	Place within 300 mm

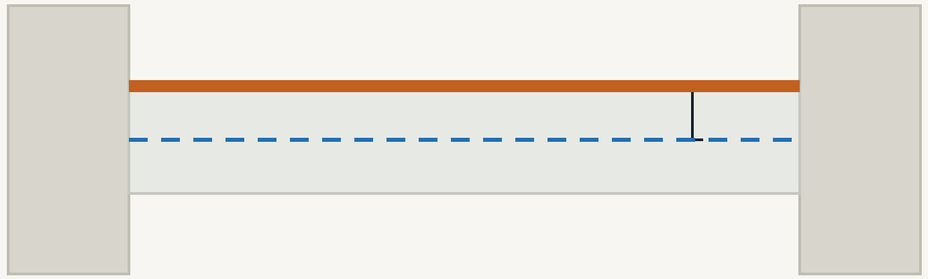
Beams: the geometry is there, the centrelines are not

Issue	What the DWG contains	What closes it
Lines are faces	Top-justified multilines, so each line is a face, 115 mm off centre	Draw one centreline per span
Mixed entity types	113 multilines, 49 plain lines and 8 polylines	One entity type for all beams
Girder widths absent	Only 2 girders are drawn, both 350 wide, against 9 GB tags from 350 to 1500	Draw girders at schedule width
Tags away from lines	B54 is 2.2 m from any beam, B87 is 2.7 m; BP1, BP2 and BP3 match nothing	Place within 300 mm
Scheduled, not tagged	B-60 appears only inside S4A/B60; B-94 and B-97 are absent; GB3A is tagged twice	One tag per span, once
Off-angle lines	The plan sits at 2 degrees, one wing at 69; stray lines run at 0, 6 and 85	Align every line to the grid
Schedule gaps	B-103 and B-106 carry no depth; GB-5 reads 1480 800/; B-34 reads 700/ and varies	Complete the schedule rows

BLOCKER 03, DRAWN OUT

Every beam in the model would sit 115 mm out

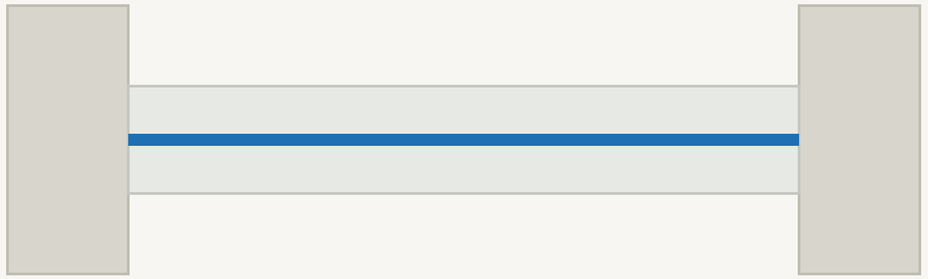
As the DWG is drawn now



Thick ochre line: the line actually in the file, lying on the beam face. The bracket is the 115 mm error.

Blue dashes: where the centreline really is.

As the script needs it



Thick blue line: one centreline per span, with multiline justification set to zero. Nothing left to guess.

A script can add 115 mm, but it cannot work out which side to add it to. On a plan rotated 2 degrees with one wing at 69, guessing the side puts beams a full width away from their columns.

Slabs and setting out are the thinnest part of the file

Slabs

No outlines to build floors from. The SLAB layer holds six open fragments. The 65 slab tags have no shape to belong to, so floors would have to be traced from beam edges or redrawn.

Sunk depths do not agree. The layers are named 300mm SUNK and 900MM SUNK, while the legend calls for 100, 150 and 400 sunk, plus the pool and deck slabs. Nobody can tell which hatch is which depth.

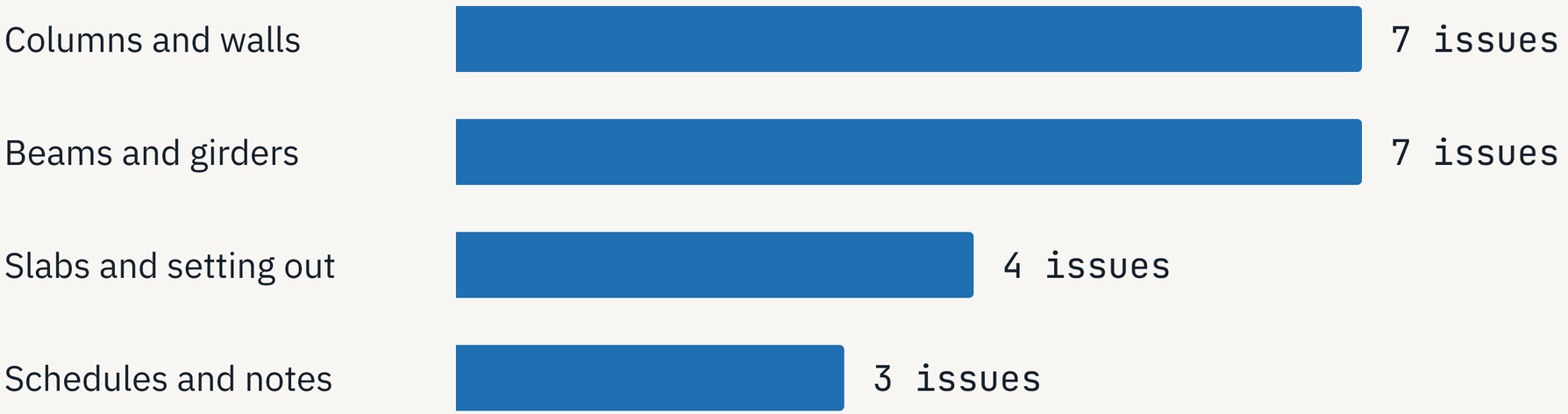
Setting out

No grid and no dimensions. Columns are numbered 1 to 32, but there is no grid to hang the model on, so the architectural grid has to stand in for it.

No shared origin. The plan sits about 2.3 km from the drawing origin, and no point is agreed with the architectural model, so nothing lines up on import.

THE AUDIT IN ONE PICTURE

21 issues, and not one of them needs new software



Every issue is traced to the rule it breaks in the standard, so the consultant receives a list of fixes rather than a list of complaints. Schedules and notes covers grouped mark rows, the M30 and M40 grade split, and the missing level table.

THE STANDARD

Structural CAD Standard for Automated Revit Modelling

01
Purpose
What the script reads, and what comes back as a query.

02
Deliverables
Per floor: DWG, schedules, levels, sections, revisions.

03
Drawing setup
Units, origin, heights, file naming, model space.

04
Layer standard
Fourteen element layers, each with its own tag layer.

05
Drawing rules
How each element is drawn, element by element.

06
Tags and schedules
Mark formats, tag placement, one row per mark.

07
Wing C findings
All 21 issues, each linked to the rule it breaks.

08
Checklist
Twelve items to tick before every issue.

Drawing setup: seven things fixed before anything is drawn

Units	Millimetres, drawn 1:1, nothing scaled or exploded
Origin	The architect's base point, identical XY on every floor, never re-rotated
Heights	Everything flat at Z = 0, with FLATTEN run before the file is issued
Files	One floor per DWG, named J1153_Wing_Level_STR_YYYY-MM-DD.dwg
Model space	The plan only, with no xrefs, no images and no viewports in it
Housekeeping	OVERKILL run, no duplicated entities, no structural elements inside blocks
Grids	Structural grid on S-GRID, matching the architectural grid line for line

STANDARD · SECTION 04

One layer per element type, and a tag layer for each

Layer	What goes on it
S-COL	Columns continuing up
S-COL-STOP	Columns stopping here
S-COL-FLOAT	Floating columns
S-WALL	Lift and pardi walls, 230
S-BEAM	Beam centrelines
S-GIRDER	Girder centrelines
S-SLAB	One outline per panel

Layer	What goes on it
S-SUNK-100/150/400	Sunk zones, by depth
S-OPEN	Openings and cut-outs
S-STAIR	Stairs and landings
S-GRID	Grid lines and bubbles
S-REF	Base point, level marks
S-ANNO	Notes, legend, title block
...-TAG	One per element layer

The current file already separates COLUMN, BEAM and GIRDER BEAM. What it lacks is a wall layer, a usable slab layer, a grid layer, and sunk layers whose names match their depth.

STANDARD · SECTION 05

How each element is drawn, so a script can trust it

Columns	One closed rectangle per column, at the scheduled size, once only
Walls	Lift walls and pardi walls on S-WALL, never on the column layer
Beams	One centreline per span, justification set to zero, parallel to the grid
Girders	Drawn at the scheduled width, not at a nominal 350 for all of them
Slabs	One closed outline per panel, edges landing on the beam centrelines
Sunk areas	A closed outline on its depth layer, not a hatch on its own
Openings	Openings, stairs and landings as closed outlines on their own layers

Marks on the plan, and the same marks in Excel

The mark formats

`C-` `FC-` `LW-` `B-` `BC-` `BP-` `MLB-` `GB-` `S-` `ST-`

- One mark per text object, never two marks in one string
- Placed inside the element, or within 300 mm of it
- No leaders, no exploded text, no marks buried in blocks

One row per mark

As issued today

```
B-2,4,7 | 230x600 | M30
```

As delivered

```
B-2 | 230x600 | M30
B-4 | 230x600 | M30
B-7 | 230x600 | M30
```

OUR SIDE OF THE WORK

A Revit template set up once, then reused on every floor

Column family	One rectangular concrete column, types named C-3 230x1200
Beam family	One concrete beam, types B 230x600 through to GB 900x800
Floor types	125, 150, 175 and 200, with sunk zones as level offsets
Wall type	230 structural wall for lift walls LW-1 to LW-4 and the pardi walls
Materials	M30 for beams and slabs, M40 for girders, as two separate materials
Shared parameters	Mark, Top Steel, Bottom Steel, Stirrups, Extra Top, Remark, Concrete Grade

With the parameters carrying the steel, the beam and column schedules come out of the model instead of being drawn by hand, and they can be checked line by line against the GFC sheet.

THE ASK

Eight things with every floor, every issue

01 The DWG, one floor per file, cleaned, flattened and set out from the agreed base point

02 Column schedule as Excel, one row per mark, numbers only

03 Beam and girder schedule as Excel, with every depth filled in

04 Slab schedule as Excel, with thickness and sunk depth per panel

05 A level table from plinth to terrace, with beam top levels

06 Sections through the sunk areas, pool, stairs and lift

07 The grid, or written confirmation that we use the architectural grid

08 A revision list naming what changed since the previous issue

The standard carries a 12-item checklist to tick before each issue

OPEN DECISIONS

Seven answers we need before the first automated run

Question	What it unblocks	Answered by
Beam lines: column face to face, or centre to centre?	Beam extraction and end trimming	Consultant
Which hatch is 100, 150 and 400 sunk? Layers say 300 and 900	Slab offsets in the wet areas	Consultant
Are C-11, 15, 16, 19 and 20 column stops?	Whether to model them at all	Consultant
C-30: is it 230x827 as drawn, or 230x950 as scheduled?	Column size and its tag	Consultant
B-60, B-94 and B-97 are scheduled but not tagged	Three beams we cannot place	Consultant
Floor-to-floor heights, plinth to terrace	Levels for every floor	Consultant
Which point is shared with the architectural model?	Setting out and alignment	Us and consultant

NEXT STEPS

Prove it on one floor, then make it the way we work

Phase 1 · [__ weeks]

Agree the standard

Issue the standard to the consultant, settle the seven open questions, and fix the shared base point with the architects.

Owner: MOFA Studio

Phase 2 · [__ weeks]

Pilot on Wing C, 1st floor

The consultant reissues this floor to the standard. We run the script, model it end to end and compare the result against the GFC sheet.

Owner: consultant, then us

Phase 3 · [__ weeks]

Roll out and write it in

Apply it to every floor and wing on J-1153, and attach the standard to the consultant's appointment for future jobs.

Owner: MOFA Studio

Durations and dates to be agreed in the meeting

Fix the drawing once. Every floor after it models itself.

Structural CAD Standard for Automated Revit Modelling, J-1153. Twenty-one findings, seven questions, one pilot floor.